Trigonometric Equation: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Types of solutions

KEY POINTS

- Principal solution: in $[0, 2\pi)$
- Particular solution: in the given interval
- General solution: all solutions by one formula, $n \in I$

2. Standard results

Formulas

$$\sin x = 0 \Rightarrow n\pi; \quad \cos x = 0 \Rightarrow (2n+1)\frac{\pi}{2}; \quad \tan x = 0 \Rightarrow n\pi$$

$$\sin x = \pm 1 \Rightarrow 2n\pi \pm \frac{\pi}{2}; \quad \cos x = 1 \Rightarrow 2n\pi; \quad \cos x = -1 \Rightarrow (2n+1)\pi$$

3. General solutions

Formulas

$$\sin x = \sin\alpha \Rightarrow x = n\pi + (-1)^n\alpha, \ \alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\cos x = \cos\alpha \Rightarrow x = 2n\pi \pm \alpha, \ \alpha \in [0, \pi]$$

$$\tan x = \tan\alpha \Rightarrow x = n\pi + \alpha, \ \alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$\sin^2 x = \sin^2\alpha$ (also $\cos^2, \tan^2$) $\Rightarrow x = n\pi \pm \alpha$

KEY POINTS

- cosec, sec, cot: solve as sin, cos, tan
- Non-standard value: $\alpha = \sin^{-1} k, \ \cos^{-1} k, \ \tan^{-1} k$
- Two forms equal: compare values in $[0, 2\pi)$ or split n even/odd

Watch out

Do not take α outside the principal interval
Both equations at once ($\tan = -1$, $\cos = \frac{1}{\sqrt{2}}$): one value in $[0, 2\pi)$, then $+2n\pi$

4. Solutions in an interval

STEPS

1. Write the general solution
2. Put $n = 0, \pm 1, \pm 2, \ldots$; keep values in the interval
3. Or count intersections of $y = \sin x$ (etc.) with $y = k$ on the graph
4. Remove points where the equation is undefined

KEY POINTS

- $\sin x = k$, $|k| < 1$: 2 solutions per 2π; $k = \pm 1$: 1 solution

5. Type 1: factorisation

STEPS

1. All terms on one side
2. Write in one ratio: quadratic in $\sin x$, $\cos x$ or $\tan x$
3. Factorise; solve each factor $= 0$
4. Keep $\sin, \cos \in [-1, 1]$, $\sin^2 \in [0, 1]$, $\tan^2 \geq 0$

Watch out

Never cancel a common factor: keep it as a factor
Parameter for which a solution exists: root of the quadratic must lie in $[-1, 1]$

6. Type 2: $a\sin x + b\cos x = c$

Formulas

$$\frac{a}{\sqrt{a^2+b^2}}\sin x + \frac{b}{\sqrt{a^2+b^2}}\cos x = \frac{c}{\sqrt{a^2+b^2}} \Rightarrow \sin(x+\theta) = \frac{c}{\sqrt{a^2+b^2}}$$

Solvable $\iff |c| \le \sqrt{a^2+b^2}$

KEY POINTS

- $\tan\theta = \frac{b}{a}$ when the value is not standard
- $t = \tan\frac{x}{2}$: quadratic; roots $\tan\frac{\alpha}{2}, \tan\frac{\beta}{2}$
- $a\sin^2 x + b\cos^2 x + c\sin x\cos x = d$: $d = d(\sin^2 + \cos^2)$, divide by $\cos^2 x$

Watch out

Dividing by $\cos^2 x$: check $\cos x = 0$ separately
$\tan\frac{x}{2}$ misses $x = (2n+1)\pi$: check it

7. Type 3: transformations

KEY POINTS

- $\sin mx = \sin nx \Rightarrow mx = k\pi + (-1)^k nx$; split k even/odd
- $\sin A = \cos B$: write $\cos B = \sin(\frac{\pi}{2} - B)$
- Sums: pair terms, use $C \pm D$, take the common factor out
- Products: $2\sin A\cos B$, $2\cos A\cos B$, ... then factorise
- $\cos x\cos(\frac{\pi}{3} - x)\cos(\frac{\pi}{3} + x) = \frac{1}{4}\cos 3x$

Watch out

$\tan mx = \tan nx$: remove values where tan is undefined

8. Type 4: substitution

Formulas

$$t = \sin x + \cos x \Rightarrow \sin 2x = t^2 - 1; \quad t = \sin x - \cos x \Rightarrow \sin 2x = 1 - t^2; \quad |t| \le \sqrt{2}$$

$$\sin^4 x + \cos^4 x = 1 - \frac{1}{2}\sin^2 2x, \quad \sin^6 x + \cos^6 x = 1 - \frac{3}{4}\sin^2 2x$$

KEY POINTS

- Other t: $\sin 2x$, $\sin^2 x$, $\tan\frac{x}{2}$, $a^{\sin^2 x}$, $\tan x + \cot x$, $\log$
- $a^{\sin^2 x} + a^{\cos^2 x} = k$: $t = a^{\sin^2 x}$, $a^{\cos^2 x} = \frac{a}{t}$

Watch out

Keep t only in its range

9. Type 5: boundedness

KEY POINTS

- $\sin^m x + \cos^n x \le 1$ for $m, n \ge 2$
- $\sin A + \sin B = 2 \Rightarrow \sin A = \sin B = 1$
- $\cos A\cos B = -1 \Rightarrow$ one is 1, the other -1
- $|a\sin x + b\cos x| \le \sqrt{a^2+b^2}$; $\tan x + \cot x \ge 2$ or ≤ -2; $2^x + 2^{-x} \ge 2$

STEPS

1. Find the range of each side
2. Equality only at max of one side = min of the other
3. Solve all conditions; take common values

10. Type 6: squaring

KEY POINTS

- Squaring adds extraneous roots
- $\sqrt{f} = g$: need $g \geq 0$ and $f \geq 0$
- $\sqrt{1 \pm \sin 2x} = |\sin x \pm \cos x|$

STEPS

1. Solve in $[0, 2\pi)$
2. Substitute each value in the original equation
3. Keep valid values, add $2n\pi$

Watch out

Domain: log needs positive argument and base $\neq 1$; $\tan, \sec$ need $\cos x \neq 0$

11. Trigonometric inequalities

STEPS

1. Reduce to one ratio of one angle: $\sin y > k$ etc.
2. Solve the equality in one period; read the inequality from the graph
3. Add $2n\pi$ ($n\pi$ for $\tan, \cot$); substitute back for x

KEY POINTS

- $\sin x > \frac{1}{2} \Rightarrow x \in (2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6})$
- Quadratic in a ratio: solve for the ratio first

Watch out

Choose the period start so the interval is not split

12. Systems of equations

KEY POINTS

- One variable: values in $[0, 2\pi)$ common to both, $+2n\pi$
- Two variables: form $x + y$, $x - y$ by adding/subtracting
- Use different integers n, m for different equations
- Ordered pairs in a box: count (x, y) for each case

13. Common mistakes checklist

STEPS

1. Cancelled a factor that can be zero?
2. Checked $\cos x \neq 0$, $\sin x \neq 0$ for $\tan, \sec, \cot, \operatorname{cosec}$?
3. Squared? Verified every root?
4. Rejected values outside $[-1, 1]$?
5. Interval ends open or closed?
6. Same integer n wrongly used for two equations?